Diploma Thesis

On the Design Space Exploration of Vibration Analysis Systems

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September, 2025

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Predictive Maintenance and Condition Monitoring

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- Alert maintenance to take action before it is to late.

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Problem Requirements

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- Actual rotating machine for deployment.

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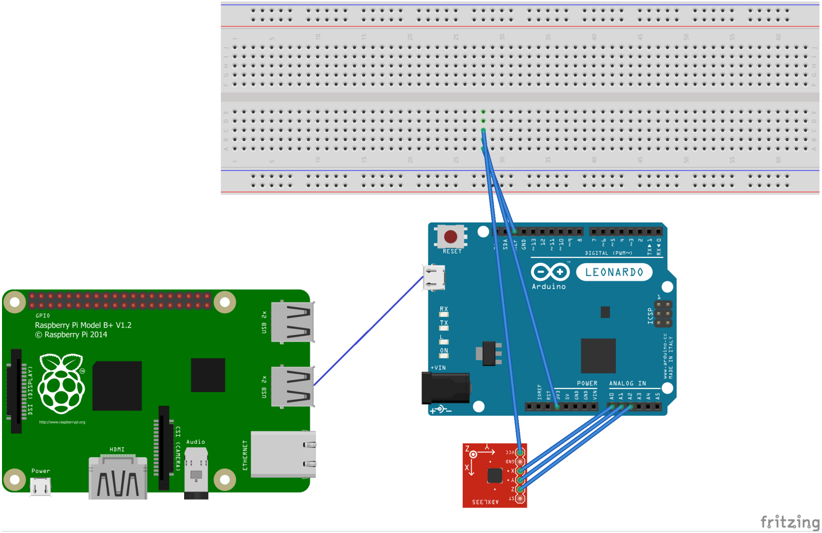
- Development of a React frontend that requests and visualizes data from the Vercel API.

Predictive Maintenance Tools



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FFT and DFT

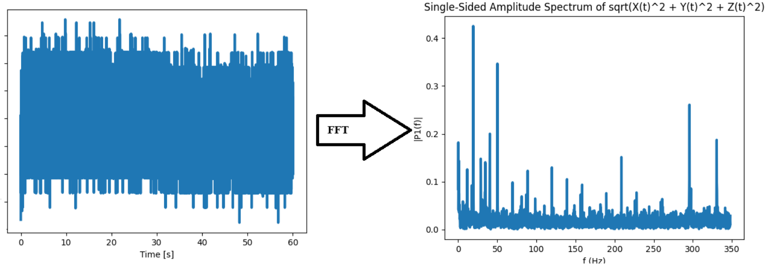

Figure 2: Conversion of the time-domain signal into its frequency-domain power representation using the FFT.

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Seeking a Repeating Pattern and Their Frequency Magnitudes

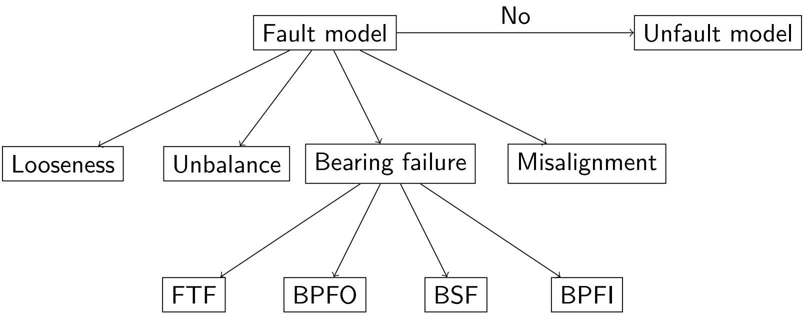

Figure 3: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

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System Architecture and Dataflow

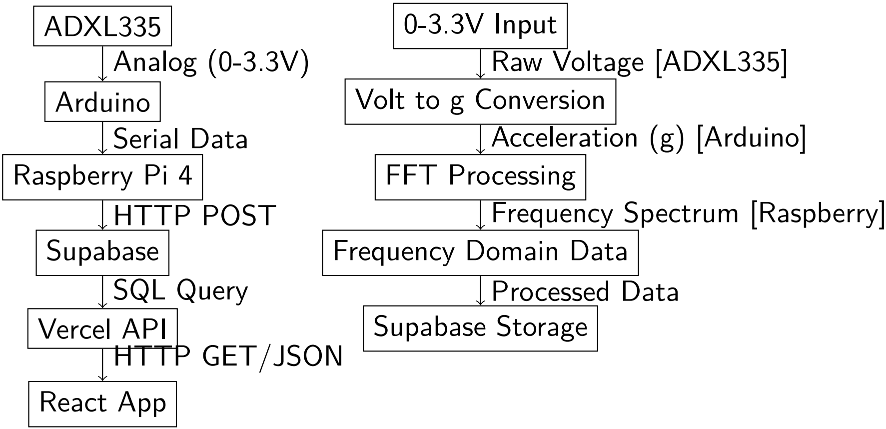

Figure 4: Complete system architecture and data flow and transformation pipeline from physical sensor to web visualization dashboard.

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- The frequency-domain analysis is limited to 350 Hz (Nyquist limit).
- The measurement time window determines the total number of samples collected.

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Arduino UNO Algorithm

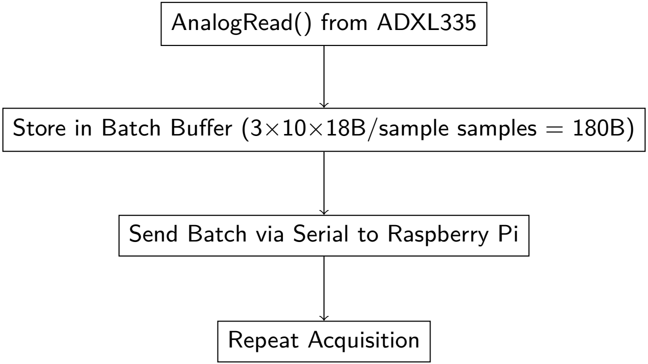

Figure 5: The Arduino UNO Algorithm.

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Raspberry Pi Algorithm

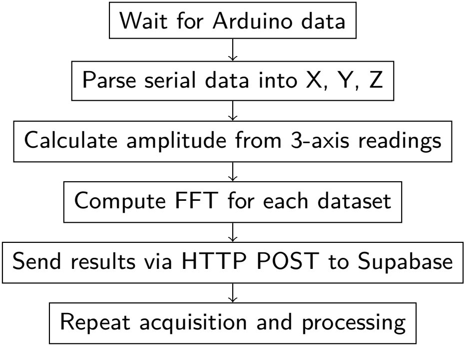

Figure 6: The Raspberry Pi 4 Algorithm: serial data reception, FFT processing, and cloud transmission.

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Vercel and Supabase Algorithmic Flow

React frontend requests vibration data via Next.js API endpoint

Next.js API receives request Uses Supabase credentials Queries database

Next.js API returns JSON response containing X, Y, Z, Magnitude measurements

React stores JSON data in local state

Extracts healthy vs current measurements Combines frequency and amplitude arrays

Generates dynamic plots using Recharts Updates plots on new data

Figure 7: Flow diagram of the Vercel and Supabase algorithm: from frontend request to dynamic visualization.

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System Verification and Validation

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- System performance validated for FFT processing, transmission rates, and visualization updates.

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Testing performance for 1[g] threshold

Table 1: Summary of critical system failures

Fault Condition	Speed	Test Cases	Correct	Incorrect Prediction
Unbalance	50 Hz	11	0	No Alarm
Misalignment	50 Hz	14	0	Correct Alarm
Ball Fault	24 Hz	3	0	No Alarm
Ball Fault	50 Hz	4	0	No Alarm
Inner Race Fault	24 Hz	4	0	No Alarm
Outer Race Fault	24 Hz	3	0	No Alarm

Testing performance for 1[g] threshold

Table 2: Summary of system test results

Task / Condition	Test Cases	Success Rate
Misalignment Prediction	14	100%
Other Bearing Failure	25	0%
Data Transmission (Raspberry $\rightarrow$ Supabase)	3	100%
Data Transmission (Arduino $\rightarrow$ Raspberry Pi)	3	100%

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Deployment on working fan for 30secs

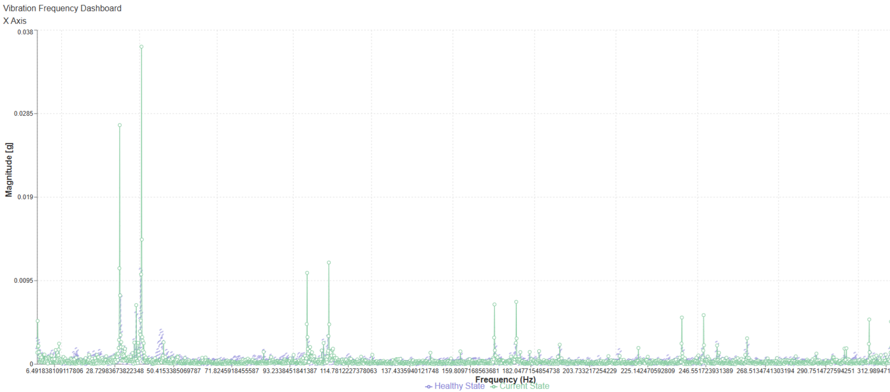

Figure 8: 30 seconds deployment, 500 frequency points and 20684 rows.

Deployment on working fan for 60secs

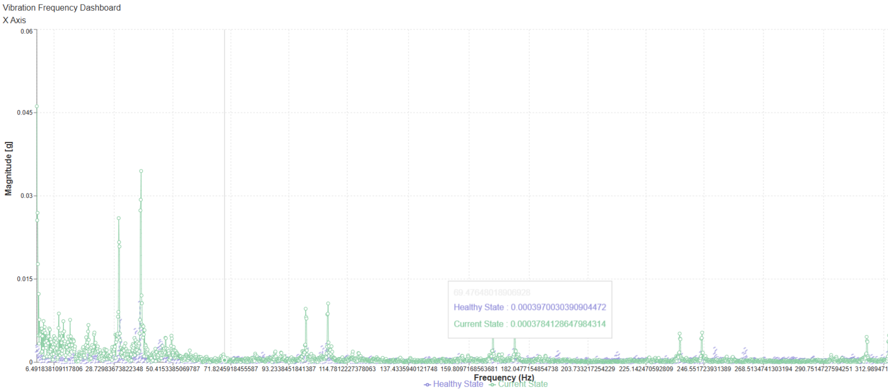

Figure 9: 60 seconds deployment, 2500 frequency points and 41434 rows.

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- Vercel dashboard visualizes frequency spectra interactively for real-time monitoring.